Resources ▼

input

Request user input

Syntax

```
x = input(prompt)
txt = input(prompt,"s")
```

Description

`x = input(prompt)` displays the text in `prompt` and waits for the user to input a value and press the **Return** key. The user can enter expressions, like `pi/4` or `rand(3)`, and can use variables in the workspace. example

- If the user presses the **Return** key without entering anything, then `input` returns an empty matrix.
- If the user enters an invalid expression at the prompt, then MATLAB® displays the relevant error message, and then redisplays the prompt.

`txt = input(prompt,"s")` returns the entered text, without evaluating the input as an expression. example

Examples

collapse all

Request Numeric Input or Expression

Request a numeric input, and then multiply the input by 10.

```
prompt = "What is the original value? ";
x = input(prompt)
y = x*10
```

At the prompt, enter a numeric value or array, such as `42`.

```
x =
    42

y =
   420
```

The `input` function also accepts expressions. For example, rerun the code.

```
prompt = "What is the original value? ";
x = input(prompt)
y = x*10
```

At the prompt, enter `magic(3)`.

```
x =
     8     1     6
     3     5     7
     4     9     2
```

```
y =
    80    10    60
    30    50    70
    40    90    20
```

Request Unprocessed Text Input

Request a simple text response that requires no evaluation.

```
prompt = "Do you want more? Y/N [Y]: ";
txt = input(prompt,"s");
if isempty(txt)
    txt = 'Y';
end
```

The `input` function returns the text exactly as typed. If the input is empty, this code assigns a default value, `'Y'`, to `txt`.

Input Arguments

collapse all

`prompt` — Text displayed to user
string | character vector

Text displayed to the user, specified as a string or character vector.

To create a prompt that spans several lines, use `\n` to indicate each new line. To include a backslash (`\`) in the prompt, use `\\`.

Output Arguments

collapse all

`x` — Result calculated from input
array

Result calculated from input, returned as an array. The type and dimensions of the array depend upon the response to the prompt.

`txt` — Exact text of input
character vector

Exact text of the input, returned as a character vector.

Algorithms

The Workspace panel does not refresh while `input` is waiting for a response from the user. Therefore, if you run `input` within a script, the Workspace panel does not display changes made to variables in the workspace until the script finishes running.

Extended Capabilities

expand all

C/C++ Code Generation
Generate C and C++ code using MATLAB® Coder™.

GPU Code Generation
Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

Version History

expand all

Introduced before R2006a

R2026a: Code generation support

See Also

`keyboard` | `inputdlg` | `listdlg` | `ginput` | `uicontrol`